Hi humans! 👋

🚀 Ready for some brainy fun?

Grid Shortest Path – Can you outsmart the maze?
Open in Streamlit

Sketch2Equation – Doodle a curve, and watch the math do its magic!
Open in Streamlit

😄 Need a giggle break?

🤣 Try my AI-powered joke machine — it might just crack you up!
Joke App Play

Curious about me?

Welcome to my GitHub profile! Here you will find a collection of projects, scripts, and code related to my professional experience in data science, software development, AI, and business analytics. I enjoy exploring various tools and technologies to solve real-world problems. Below, I provide an overview of my skills, tools, and platforms I work with.

Languages & Frameworks

VBA
Python 3.9
R
SQL
C++
Fortran
GitLab
Airflow
Vault
Confluence
Fibery
Google Apps Script

Software & Tools

MATLAB
Mathematica
SPSS
MySQL
Jupyter

Visual Studio Code

MAPLE

LaTeX

Tableau

Power BI

Microsoft Office

Salesforce

Operating Systems

Windows

Kali Linux

Databases

MySQL

DBear

My Workflow

- I leverage **VBA** and **Google Apps Script** for automating workflows and tasks in Microsoft Office and Google Workspace.
- **Python** and **R** are my go-to languages for data analysis, machine learning, and automation. I use **Jupyter Notebook** for interactive development.
- **SQL** helps me to manage and query databases, with experience in **MySQL** and **DBear** for storing and retrieving data.
- I'm experienced with version control via **GitLab** and task automation via **Airflow**.
- For reporting and visualization, I utilize **Tableau**, **Power BI**, and **Excel**.
- **Salesforce** is a platform I've integrated and developed solutions for in the business intelligence domain.

Machine Learning & Data Science Projects

Forecasting & Time Series

Temporal pattern modeling, sequential prediction, and dynamic pricing systems

Project Title	Methods & Techniques	Links
Traffic Forecasting		

Project Title	Methods & Techniques	Links
When Neural Networks Fail: A Cautionary Tale on Extrapolation Limits in Electricity Price Forecasting	• Deep Feed-Forward Neural Networks (64-64 architecture) • Relative MAE (rMAE) Custom Loss Function • Hourly Time-Series Forecasting (57,649 records) • 52-Dimensional Feature Engineering (temporal, capacity, weather, fuel prices) • Scenario Analysis Framework (2030–2050 transition pathways) • Extrapolation Limit Diagnostics (15–40x capacity growth) • Merit Order Effect Modeling • Monte Carlo Dropout for Uncertainty Quantification • Hybrid Modeling Strategy (NN + Fundamental Pricing) • Policy Target Validation (RUPTL, JETP) • MinMax/Standard Scaling Pipelines • Early Stopping & Learning Rate Scheduling	Report Code
Deep Learning for Railway Delay Prediction & Management	• Periodic Event Scheduling Problem (PESP) • Graph Neural Networks (GNN) • Graph Attention Networks (GAT) • Long Short-Term Memory (LSTM) • Spatiotemporal Modeling • Reinforcement Learning (PPO) • Delay Propagation Modeling • Uncertainty Quantification (MC Dropout) • Real-Time Prediction System	Report
Dynamic Flight Price Forecasting with XGBoost	• XGBoost Regression (500 iterations) • Feature Engineering (temporal, route, carrier) • Target Encoding • Early Stopping & Regularization • Residual Analysis • 80-20 Train-Test Split	Report

Project Title	Methods & Techniques	Links
Advanced Ensemble Flight Price Prediction (98.47% R^2)	• Stacking Ensemble • Blending Ensemble • XGBoost, LightGBM, CatBoost • Random Forest, Ridge Regression • 5-fold Cross-Validation • Meta-Learner Training	Report
Flight Price Prediction	• LightGBM with DART Boosting • Chronological Train/Test Split • SHAP Interpretability • Booking Window Simulation • Holiday Proximity Features • Revenue Elasticity Modeling	Report

🌐 Graph Machine Learning

Relational data modeling, network analysis, and graph-based recommendation

Project Title	Methods & Techniques	Links
Graph-Theoretic Analysis of Indonesian Railway Networks	• NetworkX + Neo4j Graph Database Integration • Centrality Analysis: Betweenness, PageRank, HITS (Hubs & Authorities), Degree, Closeness, Eigenvector • Vulnerability Assessment: Targeted Attack Simulation (Albert et al., 2000) • Monte Carlo Resilience Testing (50 iterations, random vs targeted) • Community Detection: Louvain Method (Modularity Q=0.769, 10 communities) • Scale-Free Network Assessment (Power-law fitting, γ=1.13) • Synthetic Dataset: 97 stations, 116 connections, 8 rail lines (KRL, MRT, LRT) • Validation against Buchwald & Sobczak (2021) Silesian Network Study • Critical Debugging Journey: HITS convergence, clustering coefficient=0, disconnected components	Report Code

Project Title	Methods & Techniques	Links
Graph-Based Dementia Detection from Clinical Speech (GraDD-ID)	• Hierarchical Graph Representation (455 nodes, 5-level ontology tree) • 348 Linguistic Features across 7 Branches (Acoustic, Discourse, Lexical, Syntactic, Psycholinguistic, Spatial, Anagraphic) • Graph Convolutional Networks (GCNN) with Global Mean Pooling • Graph2Vec + Weisfeiler-Lehman Kernel Embeddings (128-dim) • Indonesian Clinical Dataset (500 conversations, 310 patients, MMSE-labeled) • Multi-Task Learning (Binary/3-Class/5-Class Classification + MMSE Regression) • Ablation Study Framework (Branch-wise feature importance) • Cross-Validation with Statistical Significance Testing (Paired t-tests, Cohen's d) • Critical Finding: Baseline Dominance due to Dataset Constraints (500 samples insufficient for GCNN) • Scientific Integrity: Transparent documentation of graph method limitations in low-resource settings	Report Code
Deep Learning for Railway Delay Prediction & Management	• Periodic Event Scheduling Problem (PESP) • Graph Neural Networks (GNN) • Graph Attention Networks (GAT) • Long Short-Term Memory (LSTM) • Spatiotemporal Modeling • Reinforcement Learning (PPO) • Delay Propagation Modeling • Uncertainty Quantification (MC Dropout) • Real-Time Prediction System	Report
Graph-Based Fraud Detection: From Theory to Production	Industry workshop for fintech compliance teams (Feb 2026) • 45-minute technical deep dive • Live Neo4j/Python demo • Q&A on production deployment • Neo4j Aura + NetworkX • 36-client fraud ring detection • Eigenvector centrality for ringleader identification	Workshop Slides Demo Code

Project Title	Methods & Techniques	Links
Graph-Based E-Commerce Recommender System	• Neo4j Graph Database • Cypher Query Language • Jaccard Similarity Index • Collaborative Filtering • Graph Traversal Algorithms • Network Visualization • Real-time Recommendation Engine	Report Code

Optimal Transport

Project Title	Methods & Techniques	Links
Wasserstein Logistics: Optimal Transport for Indonesian Supply Chains	• Classical Kantorovich LP (Network Simplex, Birkhoff bound validation) • Entropic OT / Sinkhorn Algorithm (ε sweep 0.001–1.0) • Unbalanced OT with KL Marginal Relaxation (Chizat et al. 2018) • Partial OT with Capacity Constraints (Chapel et al. 2020) • Graph-Augmented Cost Matrices (Dijkstra shortest paths, hybrid α-interpolation) • Novel Multi-Period Dynamic OT (Temporal Frobenius coupling, λ sweep 0.01–1.0) • Block-Coordinate Proximal Sinkhorn with Gradient Clipping • 3 Indonesian Networks (Jabodetabek 12×200, Java Intercity 8×150, Archipelago 50×90) • 4 Seasonal Scenarios (Normal S/D≈1.2, Ramadan S/D≈0.4, Harbolnas S/D≈0.1, Lebaran S/D≈0.5) • Segment-Aware Demand Generation (Food ×2.5 Ramadan, Electronics ×4.5 Harbolnas) • Fairness Analysis (Gini coefficient, urban/rural fulfilment ratios up to 14:1) • N_RUNS=5 Statistical Validation with Bootstrap Confidence Intervals • POT Library (Python Optimal Transport)	Report Code

Project Title	Methods & Techniques	Links
Moving Mass, Losing Nothing: Sparse Optimal Transport via Conditional Gradient Methods	• Frank-Wolfe (Conditional Gradient) Algorithm with Exact Line Search (Jaggi, 2013) • Away-Step Frank-Wolfe with Active Set Management (Lacoste-Julien & Jaggi, 2015) • Fully Corrective Frank-Wolfe / FC-GCG (Bredies, Carioni, Fanzon & Walter, 2024) • Sinkhorn-Knopp Algorithm with Log-Domain Stabilization (Cuturi, 2013) • Quadratic-Regularized OT via Semi-Dual L-BFGS (Blondel, Seguy & Rolet, 2018) • Tsallis q-Entropy Regularization with q-Sinkhorn Iteration (Muzellec et al., 2022) • Epsilon-Scaling Sinkhorn with Cost Normalization (Schmitzer, 2019) • Wasserstein Barycenters via Iterative Bregman Projections (Cuturi & Doucet, 2014) • Sparse FW Barycenter with Alternating Plan-Barycenter Optimization • Free-Support Barycenter with Adam Optimizer • Dual Certificate Analysis & Complementary Slackness Verification • MNDSC-Related Support Gap Analysis (Carioni & Del Grande, 2023) • Linear Minimization Oracle via Network Simplex on Transportation Polytope • Indonesian Provincial Population Distribution Compression (6 Island Groups, 1,600-point grids) • 8 Experiments: Sparsity Benchmark, Pareto Frontier, Convergence Analysis, Dual Certificates, Barycenters, Indonesian Application, Scalability, Regularization Sensitivity • 6 Algorithmic Bug Fixes via Paper-by-Paper Numerical Audit (15 referenced papers)	Report Code
Bridging the Wasserstein Gap: A Systematic Empirical Validation of Optimal Transport Theory in Generative Adversarial Networks	• WGAN with Gradient Penalty • True Wasserstein Distance (LP) • 79% W_1 Reduction Achieved • Mode Collapse Elimination • Bootstrap Confidence Intervals • Statistical Significance Tests	Report Code

Project Title	Methods & Techniques	Links
Progressive Adaptive Optimal Transport for Domain Adaptation	• Four-Stage Progressive Pipeline: – Stage 1: Partial OT (80% mass transport for outlier robustness) – Stage 2: Hierarchical OT (cluster-guided cost modification, 20% reduction) – Stage 3: Low-Rank Denoising (SVD truncation, k=10) – Stage 4: Adaptive Fusion (MMD-based weighting) • Domain Adaptation Benchmarking (6 synthetic datasets: rotating_moons, partial_domain, corrupted_manifold) • Critical Failure Analysis: Transparent documentation of limitations (e.g., -11.43% degradation on gaussian_label_shift) • Statistical Validation: 5-run significance testing, MMD gap quantification • Computational Diagnostics: 6-bug debugging journey (hierarchical disaggregation disaster, API incompatibilities) • Key Result: +18.22% accuracy on partial domains where standard OT fails completely (0% → 18.22%)	Report Code

🚑 Operations Research & Public Health Optimization

Facility location, equity constrained optimization, and reinforcement learning for emergency services

Project Title	Methods & Techniques	Links
When Minutes Mean Lives: Optimizing Ambulance Placement in New York City with Equity Constraints and Reinforcement Learning	• Equity Constrained Model (ECM): Novel weighted p Median / p Center hybrid with tunable alpha parameter for efficiency equity Pareto frontier generation (Bertsimas et al., 2011) • p Median Facility Location Problem (demand weighted average RT minimization, MIP via PuLP/CBC) • Maximal Covering Location Problem / MCLP (Church & ReVelle, 1974; 8 minute threshold coverage maximization) • Maximum Expected Covering Location Problem / MEXCLP with M/M/c queueing busy fraction (Daskin, 1983) • OpenStreetMap Road Network Analysis via	Report Code Flow

Project Title	Methods & Techniques	Links
	OSMnx (Boeing, 2025): 55,268 nodes, 139,160 edges, Dijkstra shortest path OD matrix (237 x 237 ZIP codes) • Proximal Policy Optimization (PPO) for dynamic ambulance redeployment (Schulman et al., 2017; Liu & Zeng, ICLR 2024) • Hierarchical Borough Level RL Action Decomposition (Sivagnanam et al., ICML 2024) • Vectorized NumPy RL Environment (1000x faster than SimPy DES, 5 minute time steps, 288 slots/day) • Equity Aware Constrained Reward Function (Gini penalty + Rawlsian P90 penalty) • Gini Coefficient Analysis for Response Time Inequality (Enayati et al., 2023) • NYC 911 EMS Incident Dispatch Data: 18.1M cleaned incidents, 237 ZIP codes, 5 boroughs (FDNY CAD system) • Clinical Survival Estimation: OHCA decay model 7%/min (Holmen et al., 2020) • Pareto Frontier Visualization: 21 solutions across alpha sweep (0.0 to 1.0) • Key Result: ECM reduces worst case RT by 78% (1800s to 392s) with only 19% mean RT increase • RL Agent: 15.3% improvement over random redeployment (explained variance 0.957) • Estimated Impact: 93 to 299 additional cardiac arrest survivors per year • Iterative Debugging: Gini MAD linearization failure, demand scaling error, SimPy speed trap (all documented) • 20 referenced papers spanning OR, RL, health equity, and network science	

🗣 Natural Language Processing (NLP)

Text classification, sentiment analysis, and linguistic feature engineering

Project Title	Methods & Techniques	Links

Project Title	Methods & Techniques	Links
Unmasking Hoaxes in Bahasa: A Dual-Stream Knowledge-Enhanced Graph Neural Network for Indonesian Fake News Detection	• IndoBERT Indonesian Language Model (Indo4B corpus, 110M parameters) • Graph Attention Networks (GAT) with 4 heads, 2 layers • Dual-Stream Architecture (Text + Graph fusion) • Knowledge-Enhanced Entity Graphs (Type-aware edge construction) • Attention-Gated Fusion Mechanism • Focal Loss for Class Imbalance (α=0.6, γ=2.0) • 8 Indonesian News Domains (Politik, Ekonomi, Kesehatan, Teknologi, Sosial, Hukum, Pendidikan, Lingkungan) • Layer Freezing Strategy (freeze first 8 BERT layers) • Bootstrap Confidence Intervals (1,000 iterations) • McNemar's Test for Statistical Significance • 5,000-article dataset with entity annotations (120 orgs, 60 persons, 50 locations)	Report Open in Colab
Sentiment-Driven Rating Prediction for Hotel Reviews	• TF-IDF Vectorization • Count Vectorization • Random Forest (500 trees) • N-gram Analysis (1-3 grams) • 5-fold Stratified Cross-Validation • spaCy NLP Processing	Report
High-Performance Hotel Review Classification (92% Accuracy)	• Logistic Regression • TF-IDF Vectorization • Multi-class Classification (One-vs-Rest) • Text Preprocessing Pipeline • Hyperparameter Tuning (C parameter)	Report

Project Title	Methods & Techniques	Links
Comparative Analysis: ML vs Deep Learning for Sentiment	• Random Forest, SVC, Decision Tree • LSTM Neural Networks • TensorFlow/Keras • Word Embeddings (128-dim) • Dropout Regularization • GridSearchCV	Report
NLP-Based Password Strength Classification	• TF-IDF Character N-grams • Shannon Entropy Calculation • Pattern Recognition • Dictionary Word Detection • Keyboard Adjacency Detection	Report

📊 Statistical Analysis & Inference

Hypothesis testing, experimental design, and causal frameworks

Project Title	Methods & Techniques	Links
Does the Policy Actually Work? Causal Inference for Indonesian Social Protection Programs	• 13 Methods Benchmark: Classical, ML, Deep Learning, Panel Data • Classical: Propensity Score Matching (PSM), Inverse Propensity Weighting (IPW), Doubly Robust Estimation (AIPW) • Meta-Learners: S-Learner, T-Learner, X-Learner • Causal Forest & Double Machine Learning (DML) • Deep Learning: CEVAE, Counterfactual Regression Network (CFRNet) • Panel Data: Difference-in-Differences (DiD), Synthetic Control Method (SCM) • Ground Truth Evaluation: √PEHE, Coverage Rate, Monte Carlo Simulation (50 seeds) • 3 Indonesian Programs: PKH Cash Transfers, JKN Healthcare, Provincial Wage Policies • Policy-Relevant Heterogeneous Treatment Effects (HTE) • Transparent Debugging Journey: EconML discrete_treatment, CEVAE CI inconsistency	Report Code

Project Title	Methods & Techniques	Links
Predictive Validation of Multi-Level Educational Placement Systems	• Stratified Random Sampling • One-Way ANOVA & Tukey HSD • Pearson Correlation • Cohen's d Effect Sizes • Eta-Squared (η^2) • Shapiro-Wilk & Levene's Tests • Cross-Sectional Study Design	Report
Statistical Investigation of Fandango Rating Bias	• Benjamini-Hochberg FDR Correction • Bootstrap Confidence Intervals (10,000 resamples) • Benford's Law Fraud Detection • Non/Parametric Testing (Mann-Whitney U, t-tests) • Natural Experiment Design • Causal Inference Framework	Report

⚠ Anomaly Detection

Outlier identification and irregular pattern discovery

Project Title	Methods & Techniques	Links

Project Title	Methods & Techniques	Links
HybridGAD: Multi-Strategy Graph Neural Network for Financial Fraud Detection	• Graph Convolutional Networks (3-layer, 128-dim hidden) • Multi-Strategy Fusion (RQGNN + GGAD + GAD-NR) • Multi-Head Attention Mechanism (4 heads, adaptive weighting) • Financial Transaction Graph Analysis (10K-100K nodes) • Synthetic Fraud Pattern Generation (5 types: collusion, laundering, wash trading, Ponzi, camouflaged) • Barabási-Albert Scale-Free Network Topology • Class Imbalance Handling (33x fraud weighting for 2% fraud rate) • Optimal Threshold Selection via Precision-Recall Curve • Benford's Law Feature Engineering (20-dimensional financial attributes) • Early Stopping & Learning Rate Scheduling (ReduceLROnPlateau) • Camouflage Level Control (Low/Medium/High difficulty) • PyTorch Geometric Implementation (190K parameters) • Perfect Detection on Low-Camouflage Synthetic Data (F1=1.0, AUC-ROC=1.0)	Report Code

Business Analytics & Visualization

Market analysis, strategic decision support, and insight-driven visualization

Project Title	Methods & Techniques	Links
Data-Driven Market Selection for E-Learning Platform	• Market Sizing Analysis • Willingness-to-Pay Analysis • Composite Scoring • Geographic Heat Mapping • Quadrant Analysis	Report

Project Title	Methods & Techniques	Links
College Major Economic Outcomes Visualization	• Scatter Plots, Histograms, Box Plots • Distribution & Correlation Analysis • Gender Equity Analysis • Data Normalization	Report

🎤 Speaking & Workshops

Technical workshops delivered to industry audiences

Workshop Title	Audience & Context	Core Technical Project	Materials
Graph-Based Fraud Detection: From Theory to Production	Industry workshop (Feb 2026) • 45-minute technical deep dive • Live Neo4j/Python demo • Q&A on production deployment	Graph ML Project • Neo4j Aura + NetworkX • 36-client fraud ring detection • Eigenvector centrality for ringleader identification	Workshop Slides Demo Code
Graph-Based E-Commerce Recommender System	Workshop • 60-minute technical deep dive • Theoritical Fundamentals on graphs • Live Python demo	Graph ML Project • Neo4j Graph Database • Cypher Query Language • Jaccard Similarity Index • Collaborative Filtering • Graph Traversal Algorithms • Network Visualization • Real-time Recommendation Engine	Report Code